

GPU AND COOLING SYSTEM

1. Open your own laptop or desktop, identify whether it uses an integrated GPU, a discrete GPU, or both, and write down the GPU model(s) you find.

-> **Integrated GPU:** Built into the CPU/processor, commonly used for everyday tasks and light graphics.

- **Discrete GPU:** A separate graphics processor/card, generally used for gaming, 3D work and other graphics-intensive tasks.
- **Both:** Many laptops have an integrated GPU plus a discrete GPU and switch between them depending on workload.

Example: A system might show **Intel UHD Graphics + NVIDIA GeForce RTX 4050 Laptop GPU**, meaning it has both.

2. List all cooling system components you can physically see in your system (fans, heatsinks, vents, etc.), and briefly describe the function of each

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Cooling Component	Function
CPU Fan	Moves air across the CPU cooler to remove heat.
CPU Heatsink	Absorbs heat from the CPU and transfers surrounding air.
Case Fans	Move cool air into the case and hot air out.
GPU Fan	Cools the graphics processor and its heatsink.
GPU Heatsink	Removes heat generated by the GPU.
Vents / Air Intakes	Allow air to enter and leave the computer.

3. Download and use a free CPU/GPU temperature monitoring tool (like HWMonitor or Speccy), then record the idle and under-load temperatures of your CPU and GPU.

Hint: Run a browser video or simple game for 5 minutes to simulate load..

-> Procedure

Idle temperature:

1. Restart the computer.
2. Wait about 5–10 minutes without running demanding applications.
3. Open HWMonitor.
4. Record the CPU and GPU temperatures.

Load temperature:

1. Keep HWMonitor open.
2. Play a game or run a browser video for about 5 minutes.
3. Record the CPU and GPU temperatures again.
4. Create a thermal management checklist for a gaming PC, including steps for checking thermal paste, cleaning fans, and monitoring temperatures.

-> Before use

- ☐ Make sure the PC has adequate airflow.
- ☐ Check that all fans are spinning normally.
- ☐ Make sure air vents aren't blocked.
- ☐ Ensure the CPU cooler is firmly mounted.
- ☐ Check that thermal paste is properly applied.

Regular maintenance

- ☐ Turn off and unplug the PC before cleaning.
- ☐ Remove dust from fans and heatsinks.
- ☐ Clean case air filters.
- ☐ Keep the PC away from dusty or enclosed areas.
- ☐ Check CPU/GPU temperatures using monitoring software.
- ☐ Replace thermal paste when necessary, especially after removing the cooler or when temperatures become unusually high.

During gaming

- ☐ Monitor CPU/GPU temperatures.
- ☐ Check that case fans are operating.
- ☐ Investigate sudden temperature increases.

- ☐ Improve airflow if temperatures remain unusually high.
5. Given these three combinations: (A) Intel i5 CPU + GTX 1650 GPU, (B) Ryzen 9 CPU + RTX 4080 GPU, (C) Intel i3 CPU with integrated graphics, recommend an appropriate cooling solution (air or liquid) for each and justify your choice based on TDP values.

-> For a concrete comparison, I'll use representative models because “Intel i5,” “Ryzen 9,” and “Intel i3” cover multiple generations with different power characteristics.

Combination	Example CPU/GPU	Recommended Cooling	Reason
A	Intel Core i5-12400 + GTX 1650	Air cooling	Moderate power/heat; a good CPU air cooler and adequate case airflow are sufficient
B	Ryzen 9 7950X + RTX 4080	High-end liquid CPU cooling + strong case airflow	Much higher heat output; Ryzen 9 7950X has a 170 W default TDP and AMD recommends a liquid cooler for optimal performance. RTX 4080 has 320 W total graphics power.
C	Intel Core i3-12100 with integrated graphics	Basic air cooling	Lower power requirements make a basic air cooler and normal case airflow appropriate. The i3-12100 has 60 W processor base power and 89 W maximum turbo power.

A. Intel i5 + GTX 1650

For an example Core i5-12400, Intel specifies 65 W processor base power and 117 W maximum turbo power.

Recommended: Air cooling.

A quality tower air cooler plus one or more case fans provides a simple and cost-effective cooling setup.

B. Ryzen 9 + RTX 4080

The Ryzen 9 7950X has a 170 W default TDP, and AMD specifically recommends a liquid cooler for optimal performance.

The RTX 4080 has 320 W total graphics power and NVIDIA specifies a 750 W recommended system power for its reference design.

Recommended: High-end liquid CPU cooler + multiple case fans with good airflow.

The GPU itself normally uses its own built-in cooler, so the liquid cooler recommendation is primarily for managing the high-power CPU.

C. Intel i3 + Integrated Graphics

The Core i3-12100 has 60 W processor base power and 89 W maximum turbo power.

Recommended: Basic air cooling.

Because there is no separate high-power graphics card, the overall cooling requirement is relatively modest.